X Band Patch Array Antenna Design for Marine Radar Application

Mustafa Pehlivan
Electrical and Electronics Engineering
Ege University
Izmir, Turkey
mustpehl@gmail.com

Yavuz Asci
Electrical and Electronics Engineering
Ege University
Izmir, Turkey
yauuz.asci@gmail.com

Korkut Yegin
Electrical and Electronics Engineering
Ege University
Izmir, Turkey
yegink@gmail.com

Caner Ozdemir
Electrical and Electronics Engineering
Mersin University
Mersin, Turkey
cozdemir@mersin.edu.tr

***Abstract*—Horizontally polarized patch antenna array design is proposed for frequency modulated continuous wave (FMCW) marine radar applications. The array consists of series-parallel fed 96 elements printed on microwave substrate. The array configuration is optimized for minimum line loss and maximum aperture efficiency. The target frequency band is 9.3 - 9.4 GHz. Antenna parameters such as return loss, peak gain, half-power beam width and side-lobe levels are presented for marine radar applications.**

Keywords—array antenna; X band; patch array; marine radar antenna

I. Introduction

Detection and tracking of target is the main problem in radar systems [1-2]. Recently, X-band marine radars have grown because of low-cost, flexibility, high spatial and temporal resolution [3]. In radar systems microstrip patch antennas are widely used[4-5]. Radar performance depends on antenna parameters such as gain, half-power beamwidth (HPBW), side-lobe level (SLL) and polarization. Radar system needs high gain antennas [5-6]. Because of this 96 (4x24) element horizontal polarize identical microstrip antenna that are operated between 9.3-9.4 GHz array antenna is designed. For horizontally polarized has low grazing angles because of clutter, sea. It is used two common feeding mechanisms that are series and parallel in microstrip patch antenna array design. In this study, series feeding structure is used due to low loss and ease of fabrication.

II. Antenna Design

This structure is designed horizontally polarization for using marine radar application. The antenna is firstly designed as array has 4 patch elements. Antenna operates 9.4 GHz center frequency with 200 MHz bandwidth. The material of substrate has 2.55 electrical permittivity and the substrate thickness is 1.542 mm. The dimension of the patch structures are calculated and shown in the Fig. 1.

A. 1x4 Array Structure

After the patch antenna design the 1x4 array antenna structure is created to provide low HPBW at elevation. The most important point is to ensure that the signal attain at an equal phase to the antenna as seen in the Fig. 1. All patches have same sizes. But feed line widths are changed for matching.

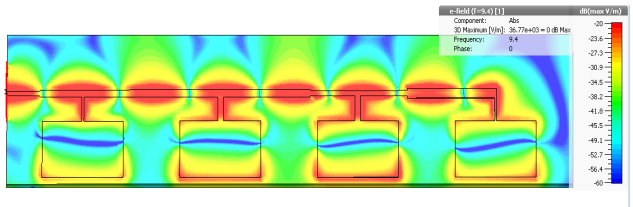

Fig. 1. E-field phase distribution on the 1x4 array structure.

The VSWR value of the 4 elements patch array is under 1.27 at 200 MHz bandwidth at center 9.4 GHz.

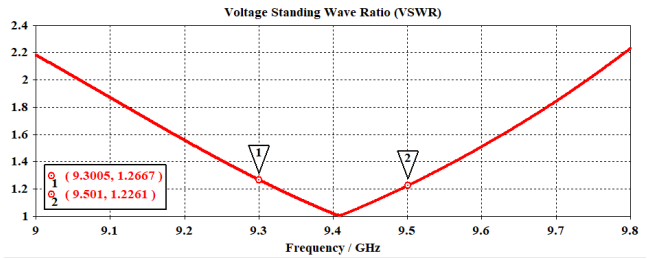

Fig. 2. VSWR value of 1x4 array structure.

When the antenna gain pattern was checked, it tilts 1 degree as shown in Fig. 3. Gain of the antenna is 10.7 dBi at 9.4 GHz and HPBW at the elevation is 18.1 degree.

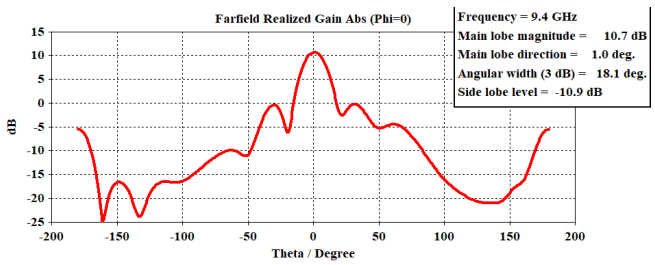

Fig. 3. Realized gain of designed 1x4 array structure

B. 24x4 Array Structure

Chebyshev algorithm is used to design feeding structure for low side lobe level array antenna. The 1x4 elements antenna is feed by feeding network that amplitude of the signals distribution of this structure are shown in Table 1. The 24x4 antenna system dimension is almost 113x485 mm.

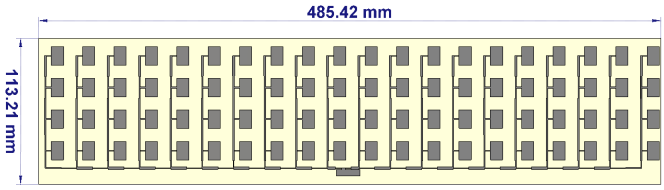

Fig. 4. 24x4 antenna array with feeding network

TABLE I. AMPLITUDE SIGNAL LEVEL AT EACH BRANCY OF FEEDING NETWORK

Port No	*Amplitude Signal Level*	Port No	*Amplitude Signal Level*
1	0.0360	13	0.2868
2	0.0425	14	0.2786
3	0.0640	15	0.2628
4	0.0902	16	0.2405
5	0.1194	17	0.2132
6	0.1508	18	0.1827
7	0.1827	19	0.1508
8	0.2132	20	0.1194
9	0.2405	21	0.0902
10	0.2628	22	0.0640
11	0.2786	23	0.0425
12	0.2868	24	0.0360

The antenna is resonating at 9.4 GHz and ensuring the desired bandwidth. The S11 parameter of the structure is shown in Fig. 5.

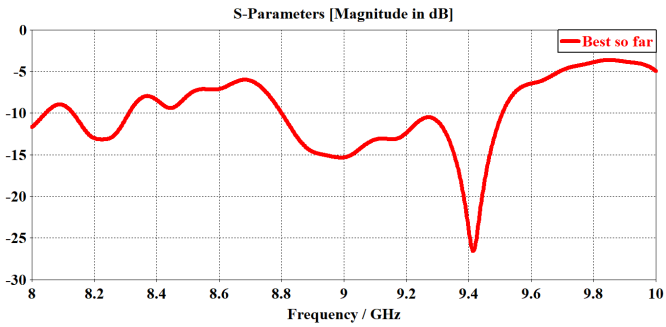

Fig. 5. S11 parameter of the 24x4 array antenna

Antenna has 23.3 dBi gains with 4.4 degree HPBW at azimuth. The SLL (Side Lobe Level) of the antenna is -38.6 dB and the back lobe level is -26.6 dB.

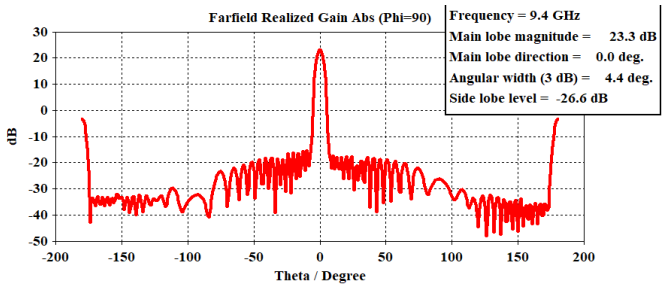

Fig. 6. Gain pattern of 24x4 array antenna structure

III. Conclusion

The horizontally polarized 4x24 elements patch array antenna is designed and simulated for using in FMCW marine radar application. It is suitable to be used in applications with the gain and side lobe level. But the back lobe level is high that expected. This situation is can be reduced using reflectors at the upper and lower part of the antenna.

IV. References

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